

Einstein-Podolsky-Rosen correlations in spontaneous parametric down-conversion: Beyond the Gaussian approximation

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We present analytic expressions for the coincidence detection probability amplitudes of photon pairs generated by spontaneous parametric down-conversion in both momentum and position spaces, without using the Gaussian approximation and taking into account the effects of birefringence in the nonlinear crystal. We also present experimental data supporting our theoretical predictions, using Einstein-Podolsky-Rosen correlations as benchmarks, for 8 different pump beam configurations.

I. INTRODUCTION

Spontaneous parametric down-conversion (SPDC) is a versatile and widely used tool in investigating fundamental quantum properties of correlated two-photon fields. Among these properties, nonclassical transverse momentum and transverse position correlations in two-photon states have been explored in many works with SPDC, in particular, the so-called Einstein-Podolsky-Rosen (EPR) paradox [1], first realized experimentally by Howell *et al.* [2] and by D'Angelo *et al.* [3].

Realizing the EPR paradox consists of preparing a quantum state of two spatially separated particles that allows one to infer with high precision either the position or the momentum of one of the particles (say, particle 1) without interacting with it, by measuring the position or the momentum of the other particle (particle 2). Since the measurement of x_2 or p_2 is a matter of choice, and x and p are incompatible variables and therefore subjected to the uncertainty relation $\Delta x \Delta p \geq \hbar/2$, EPR used the possibility to prepare such a state and the hypothesis of locality to suggest that Quantum Mechanics is an incomplete theory although it is correct in its statistical predictions. After a long debate, remarkable theoretical developments, and a long series of experiments [4], the idea that local hidden-variable theories are ruled out is now common sense. Nevertheless, EPR-type correlations are interesting on their own [5] and have been studied over the last two decades in two-photon states generated by SPDC [6–18].

As a rule, most works on EPR-type correlations in SPDC either rely on oversimplified models to describe the two-photon quantum state or do not present a theoretical model in both momentum and position representations to fit experimental data. In general, those simplified models do not include the effects of birefringence in the nonlinear crystals used in practice. Because of this, EPR correlations in SPDC have been analyzed only in

the direction that is not affected by birefringence, that is, the direction normal to the plane defined by the crystallographic optic axis and the pump beam propagation direction [19]. The commonly used Gaussian models to describe SPDC two-photon states have the advantage of simplifying calculations, but fail to correctly describe the full state propagation and do not include the effects of birefringence. A detailed discussion of Gaussian approximations in SPDC can be found in ref. [20], although in comparison with a simplified non-Gaussian model.

Experimental conditions such as the pump laser beam focusing, the nonlinear crystal length, and birefringence can affect EPR correlations in a way that is not explained by previous models. In this work, we present a more precise theoretical model for the SPDC two-photon state in both position and momentum representations and show how well it fits experimental data.

This paper is organized as follows. In Sec. II we explain the meaning of the Gaussian approximation in SPDC and present an accurate expression for the two-photon state generated by SPDC in both momentum and position representations without making use of that approximation. In Sec. III we describe EPR correlations in the two-photon states generated by SPDC and how they depend on experimental conditions. In Sec. IV we present an experiment where we measured EPR correlations in SPDC with eight different pump beam configurations, to validate our theoretical predictions. In Sec. V we present a brief discussion of the results and our conclusions.

II. THE TWO-PHOTON STATE GENERATED BY SPDC

The basic SPDC process occurs when one photon from the laser pump beam of frequency ω_p , usually in the ultra-violet spectral range, is converted into two photons of frequencies ω_1 and ω_2 , such that $\omega_1 + \omega_2 = \omega_p$ (energy conservation) and $\mathbf{k}_1 + \mathbf{k}_2 \approx \mathbf{k}_p$ (phase match). Due to dispersion, the refractive index at ω_p is greater than it is at the lower frequencies ω_1 and ω_2 , making phase match

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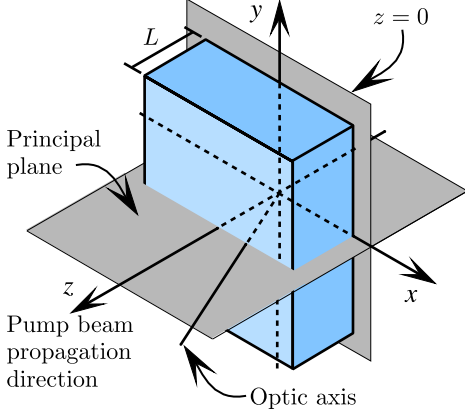


FIG. 1. Nonlinear crystal configuration.

impossible in isotropic media. This problem is circumvented in birefringent nonlinear media, where dispersion can be compensated by birefringence. For example, in negative uniaxial crystals [21], such as BBO, the pump beam polarized in the extraordinary direction and the down-converted beams polarized in the ordinary direction can be subjected to the same refractive indices, provided they propagate at appropriate directions. If the pump laser beam with extraordinary polarization propagates at an angle θ with respect to the optic axis direction inside the crystal, it is subjected to a refractive index [19]

$$\eta_p = \frac{n_{op}n_{ep}}{n_{op}^2 \sin^2 \theta + n_{ep}^2 \cos^2 \theta}, \quad (1)$$

where n_{op} and n_{ep} are the ordinary and extraordinary refractive indices, respectively, at ω_p . The collinear type I phase match condition is achieved when θ is such that, $\eta_p \omega_p = n_{o1} \omega_1 + n_{o2} \omega_2$, where n_{o1} and n_{o2} are the ordinary refractive indices at frequencies ω_1 and ω_2 , respectively.

Let us consider a piece of negative birefringent nonlinear crystal (e.g., BBO) in the form of a block having its input face lying on the plane $z = 0$, and cut for type I phase match with the principal plane (defined by optic axis and the pump beam propagation axis) parallel to the plane xz . A uv pump beam whose cross section lies entirely within the input and output faces of the crystal propagates along the z axis with extraordinary (x) polarization. The crystal thickness in the z direction is L . This configuration is illustrated in Fig. 1.

In the $\mathbf{k}$ -vector (momentum) representation, the two-photon detection probability amplitude for the state generated by SPDC in the paraxial approximation is known to be well described (up to a normalization constant) by [22]

$$\psi(\mathbf{q}_1, \mathbf{q}_2) = \mathcal{E}_0(\mathbf{q}_1 + \mathbf{q}_2) e^{-i\Delta_{oo}} \text{sinc } \Delta_{oo}, \quad (2)$$

where $\mathbf{q}_j$ is the xy component of $\mathbf{k}_j$ ($j = 1, 2$), $\mathcal{E}_0(\mathbf{q})$ is the angular spectrum of the pump beam on the plane

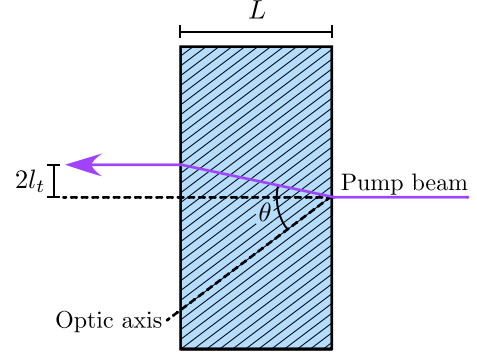


FIG. 2. The walk-off of the pump beam in a negative uniaxial crystal.

$z = 0$,

$$\Delta_{oo} = \mu_{oo} + l_t(q_{1x} + q_{2x}) - \frac{L}{4k_p} \left| \sqrt{\frac{\omega_2}{\omega_1}} \mathbf{q}_1 - \sqrt{\frac{\omega_1}{\omega_2}} \mathbf{q}_2 \right|^2,$$

$k_p = \eta_p \omega_p / c$, l_t is half of the transverse walk-off length, $\mu_{oo} = (\bar{n}_o - \eta_p) k_p L / 2 \eta_p$, and $\bar{n}_o$ is the ordinary refractive index of the nonlinear crystal at $\omega_p/2$. The transverse walk-off length is given by [19]

$$2l_t = \frac{(n_{op}^2 - n_{ep}^2) \sin \theta \cos \theta}{n_{op}^2 \sin^2 \theta + n_{ep}^2 \cos^2 \theta} L. \quad (3)$$

In negative uniaxial crystals, $n_{ep} < n_{op}$ and the extraordinary beam tends to deviate away from the optic axis direction. This situation is illustrated in Fig. 2

Here we work in the collinear phase match condition, where $\bar{n}_o = \eta_p \rightarrow \mu_{oo} = 0$, and in the quasi-degenerate regime, where $\omega_1 = (1 + \nu) \omega_p / 2$ and $\omega_2 = (1 - \nu) \omega_p / 2$, with $\nu \ll 1$. This means that $\sqrt{\omega_2 / \omega_1} \approx 1 - \nu$ and $\sqrt{\omega_1 / \omega_2} \approx 1 + \nu$. In the collinear phase match it is possible to make a straightforward use of the paraxial approximation and Fourier optics [23]. It is also possible to do so in the noncollinear phase match but at the cost of more complicated expressions. We chose to work in the quasi-degenerate regime in order to stress the generality of our model. The degenerate regime can be readily recovered by making $\nu = 0$ in the expressions that follow. It is interesting to notice that even in the collinear regime there will be a walk-off in the propagation of the pump beam inside the nonlinear crystal, since the phase match angle θ lies, in general, between 0 and $\pi/2$. In our experiment (see section IV), $\theta \approx 33^\circ$.

To simplify the notation, we make

$$\begin{aligned} \mathbf{Q} &= \mathbf{q}_1 + \mathbf{q}_2, \\ \mathbf{P} &= (1 - \nu) \mathbf{q}_1 - (1 + \nu) \mathbf{q}_2, \\ \beta^2 &= \frac{L}{4k_p}. \end{aligned} \quad (4)$$

Then,

$$\psi(\mathbf{Q}, \mathbf{P}) = \mathcal{E}_0(\mathbf{Q}) \text{sinc}(l_t Q_x - \beta^2 P^2) e^{-i(l_t Q_x - \beta^2 P^2)}. \quad (5)$$

Since $\psi(\mathbf{Q}, \mathbf{P})$ is a two-beam plane-wave spectrum and the two beams propagate independently by acquiring a phase factor depending on the z -component of each $\mathbf{k}$ -vector [23], on the output face of the nonlinear crystal, at $z = L$, the propagated $\psi(\mathbf{Q}, \mathbf{P})$ is

$$\psi(\mathbf{Q}, \mathbf{P}, L) = \psi(\mathbf{Q}, \mathbf{P}) e^{i(k_{1z} + k_{2z})L}. \quad (6)$$

Inside the crystal, in the collinear phase match [22],

$$\begin{aligned} (k_{1z} + k_{2z})L &= \frac{\bar{n}_o L}{c} (\omega_1 + \omega_2) \\ &- \frac{Lc}{\bar{n}_o} [(1 - \nu)q_1^2 + (1 + \nu)q_2^2] \\ &= k_p L - 2\beta^2(Q^2 + P^2). \end{aligned} \quad (7)$$

Therefore,

$$\begin{aligned} \psi(\mathbf{Q}, \mathbf{P}, L) &= \mathcal{E}_0(\mathbf{Q}) \text{sinc}(l_t Q_x - \beta^2 P^2) \\ &\times e^{-i(l_t Q_x + 2\beta^2 Q^2)} e^{-i\beta^2 P^2}. \end{aligned} \quad (8)$$

Considering a Gaussian pump beam, we can write, in the complex notation [24],

$$\mathcal{E}(\mathbf{Q}, z) = A e^{ik_p^v z} e^{-ia(z)Q^2/2k_p^v}, \quad (9)$$

where A is a constant, $k_p^v = \omega_p/c$ (superscript v stands for vacuum), $a(z) = z - z_c - ik_p^v w_0^2/2$, w_0 is the beam waist and z_c is the waist location on the z axis. Hence,

$$\mathcal{E}_0(\mathbf{Q}) = A e^{-ia_0 Q^2/2k_p^v}, \quad (10)$$

where $a_0 = a(0)$. Then, at the output face of the crystal,

$$\begin{aligned} \psi(\mathbf{Q}, \mathbf{P}, L) &= e^{-i(a_0/2k_p^v + 2\beta^2)Q^2} e^{-il_t Q_x} e^{-i\beta^2 P^2} \\ &\times \text{sinc}(l_t Q_x - \beta^2 P^2). \end{aligned} \quad (11)$$

From $z = L$ to $z > L$, $\psi(\mathbf{Q}, \mathbf{P})$ propagates in free space, that is,

$$\psi(\mathbf{Q}, \mathbf{P}, z) = \psi(\mathbf{Q}, \mathbf{P}, L) e^{i(k_{1z}^v + k_{2z}^v)(z-L)}. \quad (12)$$

In free space and collinear propagation,

$$\begin{aligned} (k_{1z}^v + k_{2z}^v)(z - L) &= \left(k_1^v + k_2^v - \frac{q_1^2}{2k_1^v} - \frac{q_2^2}{2k_2^v}\right)(z - L) \\ &= k_p^v(z - L) + 2\bar{n}_o\beta^2(Q^2 + P^2) \\ &- \frac{z}{2k_p^v}(Q^2 + P^2). \end{aligned} \quad (13)$$

Therefore,

$$\begin{aligned} \psi(\mathbf{Q}, \mathbf{P}, z) &= \text{sinc}(l_t Q_x - \beta^2 P^2) e^{-il_t Q_x} \\ &\times e^{-ib_1(z)Q^2} e^{-ib_2(z)P^2}, \end{aligned} \quad (14)$$

where

$$\begin{aligned} b_1(z) &= a(z)/2k_p^v - 2(\bar{n}_o - 1)\beta^2, \\ b_2(z) &= z/2k_p^v - (2\bar{n}_o - 1)\beta^2. \end{aligned} \quad (15)$$

To the best of our knowledge, an accurate analytic expression for ψ in the coordinate representation is not available in the literature, except under the Gaussian approximation, which consists of replacing $\text{sinc}(l_t Q_x - \beta^2 P^2)$ in Eq. (14) by a Gaussian function, necessarily neglecting the walk-off term $l_t Q_x$. This approximation works well when $L/k_p w_0^2 \ll 1$, that is, when the pump beam is highly collimated, or when the crystal is very thin ($L \approx 1$ mm) [20].

To arrive at an expression for ψ , in coordinate representation, we adopt the following approximation: For pump beams with reasonably narrow-band angular spectra ($w_0 \geq 50 \mu\text{m}$) and $L \sim 1$ mm to 5 mm, which cover most practical cases, we can make

$$\begin{aligned} \psi(\mathbf{Q}, \mathbf{P}, z) &= \text{sinc}(l_t Q_x) e^{-il_t Q_x} e^{-ib_1(z)Q^2} \\ &\times \text{sinc}(\beta^2 P^2) e^{-ib_2(z)P^2}, \end{aligned} \quad (16)$$

which turns $\psi(\mathbf{Q}, \mathbf{P}, z)$ into a separable function of $\mathbf{Q}$ and $\mathbf{P}$.

Defining the coordinates $\mathbf{R} = [(1 + \nu)\boldsymbol{\rho}_1 + (1 - \nu)\boldsymbol{\rho}_2]/2$ and $\mathbf{S} = (\boldsymbol{\rho}_1 - \boldsymbol{\rho}_2)/2$, the calculation of the Fourier transform of $\psi(\mathbf{Q}, \mathbf{P}, z)$ is straightforward:

$$\begin{aligned} \psi(\mathbf{R}, \mathbf{S}, z) &= \frac{e^{-R_y^2/4ib_1(z)}}{l_t \sqrt{ib_1(z)}} \\ &\times \left\{ \text{Erf} \left[\frac{R_x - 2l_t}{2\sqrt{ib_1(z)}} \right] - \text{Erf} \left[\frac{R_x}{2\sqrt{ib_1(z)}} \right] \right\} \\ &\times \left\{ \text{Ei} \left[\frac{ik_p^v S^2}{2(z - L)} \right] - \text{Ei} \left[\frac{ik_p^v S^2}{2(z - L')} \right] \right\} \end{aligned} \quad (17)$$

for $z > L$, where $L' = (1 - 1/\bar{n}_o)L$, Erf is the error function, and Ei is the exponential integral function [25].

Eqs. (16) and (17) can be considered the main contribution of this work to the field of SPDC. Using Eq. (17) one can predict experimental results of spatial two-photon correlations with very good accuracy and push experimental conditions to their limits when testing specific theoretical models. In this work, we test EPR correlations, as they involve both spatial and momentum correlations.

III. EPR CORRELATIONS

With expressions (16) and (17) we can calculate the uncertainties Δx_1 , Δy_1 , for fixed x_2 , y_2 and z , and Δk_{x1} , Δk_{y1} for fixed k_{x2} , k_{y2} as functions of the pump beam angular spectrum width (defined by the beam waist w_0) and see how they are affected by the crystal anisotropy (quantified here by walk-off parameter l_t). Assuming any fixed value for $\boldsymbol{\rho}_2$, Eq. (17) allows us to calculate Δx_1